

Notes: Rotemberg and Saloner (AER, 1994)

Benefits of Narrow Business Strategies

Contents

1	One-sentence summary	2
2	Big picture	2
2.1	Question	2
2.2	Core answer	2
2.3	Main friction	2
2.4	Main mechanism	2
2.5	Economic takeaway	2
3	Incremental contribution of the paper	2
3.1	A formal incentive theory of narrow strategy	2
3.2	Synergy can be harmful	3
3.3	Incomplete contracts as the key driver	3
3.4	Organizational scope and internal capital markets	3
3.5	Bridge to diversification discount evidence	3
4	Conceptual framework	3
4.1	Innovation and implementation	3
4.2	Implementation-contingent compensation	3
4.3	Narrowness as commitment	4
5	Model setup and derivations	4
5.1	One activity: setting 1	4
5.2	One activity: setting 2	4
5.3	Lemma 2: three incentive cases	5
6	Main cases in the paper	5
6.1	Case 1: synergy as a substitute for the core innovation	5
6.2	A's value in a broad firm	5
6.3	Ex post and ex ante inefficiency	5
6.4	Proposition 2: narrowness can be profitable	6
6.5	Case 2: financial constraints and internal capital competition	6
6.6	Financial constraints and asymmetry	6
7	Economic intuition	6
7.1	Why broadness can destroy incentives	6
7.2	Why simple diversification logic fails	6

7.3	Why relatedness is ambiguous	6
7.4	Why large scale differs from broad scope	7
7.5	Organizational remedies	7
8	Identification and interpretation	7
8.1	What the paper identifies theoretically	7
8.2	Predictions and implications	7
8.3	Limitations	7
9	Tables and figures	7
9.1	Figure 1: Ex Ante and Ex Post Inefficiency; Notes-created Table: Reading the Regions	8
9.2	Notes-created Tables: Core Model Objects and Main Results	9
10	Short summary	9
11	Conclusion	9
11.1	Core conclusion	9
11.2	Economic intuition	9
11.3	Incremental contribution revisited	10
11.4	Relation to diversification discount	10
11.5	Limitations	10
11.6	Takeaway	10

1 One-sentence summary

Rotemberg and Saloner explain why innovation-oriented firms may rationally pursue narrow business strategies: narrowness can strengthen employee innovation incentives when incomplete contracts make broad firms prone to ex post idea substitution or internal capital competition.

2 Big picture

2.1 Question

Why do firms sometimes reject apparently profitable opportunities that do not fit their core strategy, even though standard profit maximization seems to recommend taking every positive-NPV project?

2.2 Core answer

Narrow strategy can be valuable because it commits the firm not to use noncore opportunities as substitutes for employees' core innovations. This commitment improves employees' ex ante incentives to search for improvements in core activities.

2.3 Main friction

The firm cannot write complete contracts on employee effort or on the exact value of an idea. It can only reward an employee when the employee's idea is implemented. This implementation-contingent incentive contract distorts ex post implementation choices.

2.4 Main mechanism

A broad firm has more alternative projects or more substitute ideas. When an employee knows that a substitute innovation may displace his own idea, his probability of receiving the implementation bonus falls. If this effect is strong, he will not exert costly effort. A narrow firm avoids this problem by excluding the substitute activity.

2.5 Economic takeaway

Diversification can lower value not merely because managers are distracted or because activities lack synergies. It can lower value precisely because synergies and internal capital markets change ex post implementation incentives and weaken ex ante innovation effort.

3 Incremental contribution of the paper

3.1 A formal incentive theory of narrow strategy

The paper gives an economic foundation for a common strategy recommendation: firms should have narrow, clearly defined objectives. It explains narrowness as an incentive device, not simply as focus, specialization, or managerial taste.

3.2 Synergy can be harmful

The paper makes the counterintuitive point that synergies between activities can reduce firm value. Synergies are harmful when they create ex post substitutes for core innovations and thereby reduce ex ante effort incentives.

3.3 Incomplete contracts as the key driver

The result does not rely on irrational managers, fixed attention, or technological diseconomies of scope. It relies on incomplete contracting: the firm cannot make incentive pay perfectly contingent on effort or on the exact quality of the innovation.

3.4 Organizational scope and internal capital markets

The paper connects corporate scope with financial constraints. A broad firm may create an internal capital market, but when funds are scarce, projects in noncore activities can compete with core projects and weaken the incentives of core employees.

3.5 Bridge to diversification discount evidence

The paper offers a mechanism consistent with evidence that unrelated diversification can reduce firm value: diversification can undermine innovation incentives by expanding the set of projects that senior management can implement instead of the employee's innovation.

4 Conceptual framework

4.1 Innovation and implementation

The model separates two steps:

- **Discovery:** an employee exerts costly effort to find a profit-improving idea.
- **Implementation:** senior management decides whether to implement the idea.

The key assumption is that the employee can be rewarded only if his idea is implemented.

Interpretation: This separation is essential. Even if an innovation is valuable, the employee may not expect to be paid if management later finds a substitute project that is even more attractive or cheaper to implement.

4.2 Implementation-contingent compensation

A worker must receive a minimum wage w . If his idea is implemented, he can receive an additional incentive payment k . The payment cannot be perfectly conditioned on effort or on the exact value of the idea.

Interpretation: The incentive contract is coarse. The firm can pay for implementation, but cannot pay for effort or for the discovery of an idea that management later chooses not to use.

4.3 Narrowness as commitment

A narrow firm removes alternative projects from management's future choice set. This can be beneficial even if the excluded activity is profitable on its own, because the exclusion reassures core employees that their ideas will not be displaced.

Interpretation: Narrowness is an organizational commitment technology. It reduces ex post flexibility but can raise ex ante effort.

5 Model setup and derivations

5.1 One activity: setting 1

The firm operates one activity, a . If the employee exerts effort at cost d , then with probability Q the firm can implement an idea that raises revenues from S to $S + g$.

The effort is socially worthwhile if

$$Qg - d > 0. \quad (1)$$

A first-best incentive contract sets

$$k = \frac{d}{Q}. \quad (2)$$

Interpretation: If the employee succeeds, the idea is valuable enough to implement even after paying k . The firm therefore implements every successful idea, and the worker is exactly compensated for effort in expectation.

5.2 One activity: setting 2

Now the employee's effort can produce two positive outcomes:

$$S + g \quad \text{with probability } Q', \quad S + z \quad \text{with probability } Q'', \quad S \quad \text{otherwise,} \quad (3)$$

with $0 < z < g$ and

$$Q'g + Q''z = Qg. \quad (4)$$

Thus the expected social value of effort is the same as in setting 1, but the distribution of outcomes is more intermediate and less jackpot-like.

The paper's two key conditions are

$$z \geq \frac{d}{Q' + Q''}, \quad (5)$$

$$g \geq \frac{d}{Q'}. \quad (6)$$

Interpretation: A low incentive payment can motivate effort only if the firm implements all successful ideas, including moderate ones. A high incentive payment can motivate effort only if the high payoff state is sufficiently likely. If neither condition holds, the firm cannot elicit effort.

5.3 Lemma 2: three incentive cases

- If $z \geq d/(Q' + Q'')$, the firm can implement all positive ideas and achieve the first best.
- If $z < d/(Q' + Q'')$ but $g \geq d/Q'$, the firm can motivate effort only by paying enough to implement the high-value idea. Moderate-value ideas are wasted; this is ex post inefficient.
- If both conditions fail, no implementation-contingent payment elicits effort; this is ex ante inefficient.

Detailed interpretation: The model shows why a richer set of possible project outcomes can make incentives harder. A moderate idea may be socially valuable but not worth implementing after paying the bonus. If the worker expects many such states, he will not exert effort.

6 Main cases in the paper

6.1 Case 1: synergy as a substitute for the core innovation

The broad firm adds a second activity, b . Employee B can discover an innovation for b . If B succeeds, his idea may also be useful in a with probability $\bar{P}$, generating value $\bar{g}$ in activity a .

If A succeeds in activity a , his idea is worth g_a . But if B also succeeds and his idea can be applied in a , management may implement B 's idea instead.

Interpretation: The synergy is valuable in a complete-contract world, but it reduces the value of A 's effort under incomplete contracts because it creates a substitute for A 's idea.

6.2 A's value in a broad firm

The value of A 's effort in the broad firm has three states:

$$\begin{aligned} g_a, & \quad \text{with probability } P_a(1 - P_b\bar{P}), \\ g_a - \bar{g}, & \quad \text{with probability } P_aP_b\bar{P}, \\ 0, & \quad \text{with probability } 1 - P_a. \end{aligned} \tag{7}$$

The effort remains socially worthwhile if

$$P_ag_a - P_aP_b\bar{P}\bar{g} - d_a > 0. \tag{8}$$

Interpretation: Breadth turns some of A 's successes into intermediate outcomes. When the intermediate payoff is too low to justify paying the incentive bonus, A 's motivation can unravel.

6.3 Ex post and ex ante inefficiency

The two critical inequalities are

$$g_a - \frac{d_a}{P_a} - \bar{g} \geq 0, \tag{9}$$

$$g_a - \frac{d_a}{P_a(1 - P_b\bar{P})} \geq 0. \tag{10}$$

When the first fails but the second holds, effort can be elicited but the firm sometimes discards A 's socially valuable innovation. When both fail, A cannot be motivated to exert effort.

Interpretation: The figure in the paper maps exactly these regions. The vertically shaded area is ex ante inefficiency; the horizontally shaded area is ex post inefficiency.

6.4 Proposition 2: narrowness can be profitable

A narrow firm that focuses only on activity a gives up the direct profits from b , but it also avoids the incentive destruction caused by B 's possible substitute innovation. The paper shows that there are parameter regions in which this trade-off favors narrowness.

Economic message: Narrowness is profitable not because b is unprofitable, but because the presence of b changes how hard it is to motivate innovation in a .

6.5 Case 2: financial constraints and internal capital competition

The paper then studies a firm that cannot borrow to finance implementation. If both A and B produce ideas, the firm has internal cash to finance only one. Thus the projects compete for scarce internal funds.

Interpretation: A broad firm's internal capital market can undermine incentives. Employees understand that even if they succeed, their projects may lose the internal funding contest.

6.6 Financial constraints and asymmetry

Assume a is more valuable ex ante:

$$P_a g_a - d_a \geq P_b g_b - d_b. \quad (11)$$

The firm could still benefit from b if it were stand-alone. But if b sometimes displaces a 's project when both succeed, then A may not exert effort. The paper shows that narrow focus on a can again dominate.

Interpretation: The financial-constraints model shows that narrowness is valuable even without technological synergy. Internal funding competition alone can create the same incentive problem.

7 Economic intuition

7.1 Why broadness can destroy incentives

An employee's incentive depends on the probability that his idea will actually be implemented. Broadness gives management more alternatives. These alternatives are valuable ex post but reduce the employee's expected implementation probability ex ante.

7.2 Why simple diversification logic fails

Standard diversification logic counts the direct payoff from adding a new activity. Rotemberg and Saloner add an incentive cost: the new activity changes the internal allocation of attention, implementation rights, and capital.

7.3 Why relatedness is ambiguous

Related activities can create synergies, which is good. But they can also create substitutes for core innovations, which is bad. Related diversification is therefore not always better than unrelated diversification.

7.4 Why large scale differs from broad scope

Large scale can increase the value of an employee's innovation without necessarily creating substitute projects. Broad scope creates additional project options that may displace the employee's idea.

7.5 Organizational remedies

The paper discusses mechanisms such as Chinese walls, separate divisions, or restrictions on internal capital flows. These tools reduce the ability of one activity to supplant another, approximating the commitment value of narrowness.

8 Identification and interpretation

8.1 What the paper identifies theoretically

This is a theoretical paper, not a causal empirical design. It identifies a mechanism: incomplete contracts plus implementation-contingent incentives can make broad organizational scope privately unprofitable.

8.2 Predictions and implications

The model predicts that narrow strategies are most useful when:

- innovation effort is important;
- employees are paid only when ideas are implemented;
- broad scope creates substitute projects or internal funding competition;
- employees cannot be credibly promised implementation priority;
- activities are asymmetric, so one activity is core and the other threatens to supplant it.

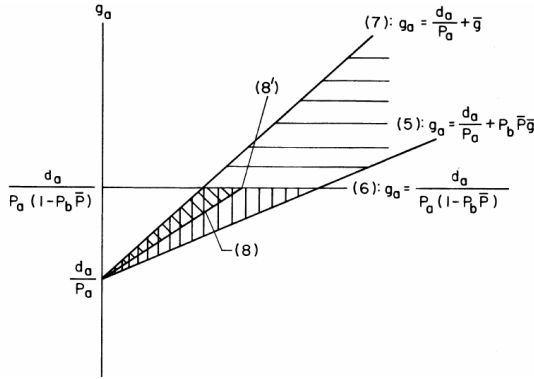
8.3 Limitations

The paper abstracts from explicit managerial agency problems, formal capital-budgeting procedures, multiple layers of hierarchy, dynamic learning, and empirical measurement. It provides a mechanism rather than a quantitative estimate.

9 Tables and figures

The original article has one main figure and no empirical tables. To keep the notes compact, Figure 1 is paired with a notes-created table that summarizes the regions in the figure.

9.1 Figure 1: Ex Ante and Ex Post Inefficiency; Notes-created Table: Reading the Regions



(a) **Figure 1: Ex Ante and Ex Post Inefficiency.** The vertical axis is g_a and the horizontal axis is $\bar{g}$. The lines correspond to inequalities (5), (6), (7), (8), and (8'). The shaded regions show where broadness creates ex ante or ex post inefficiency.

Region/line	Economic meaning
(5)	A's effort is socially worthwhile after accounting for possible displacement by B's idea.
(6)	A's idea remains worth implementing even when B has a substitute idea.
(7)	A can be motivated even when his bonus is paid only in states where B does not displace him.
Vertical shading	Ex ante inefficiency: A cannot be motivated to exert effort.
Horizontal shading	Ex post inefficiency: A works, but the firm sometimes implements the wrong substitute.
Cross-hatching	Narrowness can dominate broadness.

(b) **Notes-created table: Reading Figure 1.** The figure maps parameter regions in which adding a synergistic activity undermines incentives.

Figure 1 plus interpretation table. The paired layout places the original figure and its reading guide on the same row, following the requested compact format.

Read: The lower sloping line defines when A's effort is socially worthwhile. The upper and middle lines separate regions where A can be motivated, where implementation is inefficient, and where effort collapses. The cross-hatched region is the key one: a broad firm has access to B's idea, but that access makes it less profitable than a narrow firm focused on a .

Economic message: The figure visualizes the paper's central insight. Breadth is not always good even when it creates synergies, because the very availability of synergistic substitutes changes incentives in the core business.

Detailed interpretation: If $\bar{g}$ is small, the synergy from B does little to displace A, so broadness may be beneficial. If $\bar{g}$ is intermediate, B's idea is a powerful enough substitute to undermine A's incentives, making narrowness attractive. If $\bar{g}$ is very large, abandoning A may be efficient even for the planner, so broadness can again be attractive.

9.2 Notes-created Tables: Core Model Objects and Main Results

Object	Role in model
A, B	Employees who can generate ideas in activities a, b .
d_a, d_b	Effort costs, paid privately by employees.
P_a, P_b	Probabilities that employees successfully generate ideas.
g_a, g_b	Net payoff from implementing own-activity innovations.
$\bar{P}, \bar{g}$	Probability and value that B 's idea is also useful in a .
k	Implementation-contingent incentive payment.

(a) **Notes-created table: Model primitives.** These primitives organize both the synergy and financial-constraints versions of the model.

Result	Interpretation
Lemma 1	In the one-outcome setting, implementation bonuses can attain first best.
Lemma 2	With intermediate outcomes, incentives can generate ex post inefficiency or fail entirely.
Prop. 1	Broadness maps the core activity into the intermediate-outcome problem.
Prop. 2	Narrowness can be strictly more profitable than adding a profitable activity.
Props. 3-4	More complex contracts may reduce but not always eliminate the problem.
Props. 5-8	Financial constraints and internal capital competition create a parallel rationale for narrowness.

(b) **Notes-created table: Main theoretical results.** The propositions show how incomplete contracts, synergies, and financial constraints interact with organizational scope.

Model objects and results. These notes-created tables are paired to keep the theoretical summary compact.

10 Short summary

- The paper explains why innovation-oriented firms may rationally pursue narrow strategies.
- The mechanism is incomplete contracting: employees can be rewarded only if their ideas are implemented.
- Broad scope gives management more substitute projects, reducing employees' expected probability of being rewarded.
- Synergies can be harmful when they make one employee's idea a substitute for another's.
- Narrowness can be a commitment not to supplant core employees' innovations.
- Financial constraints produce a similar mechanism through competition for internal funds.
- The value of narrowness is strongest when activities are asymmetric and core innovation effort is important.
- The theory helps explain the diversification discount and firms' insistence on sharply defined strategic goals.

11 Conclusion

11.1 Core conclusion

Rotemberg and Saloner show that firms can benefit from narrow strategies because narrowness helps motivate employees to search for innovations. A broad firm may have more opportunities, but those opportunities can undermine the implementation-contingent incentives needed to induce core employees to exert effort.

11.2 Economic intuition

The key economic intuition is a trade-off between ex post flexibility and ex ante incentives. Ex post, management likes having many projects and substitute ideas. Ex ante, employees dislike the fact that their

ideas may be displaced. If contracts cannot reward effort or idea discovery directly, narrowing the firm can be the only credible way to protect employee incentives.

11.3 Incremental contribution revisited

The paper provides a theoretical foundation for business-strategy advice that economists might otherwise dismiss as vague. “Stick to the core” is not merely managerial rhetoric. In environments with incomplete contracts and innovation incentives, it can be an efficiency-enhancing commitment.

11.4 Relation to diversification discount

The theory is consistent with evidence that unrelated diversification can reduce profitability and market value. It also predicts that broadness is most damaging for innovative firms, because their profits depend heavily on employee idea generation.

11.5 Limitations

The model is stylized. It does not quantify how often firms face this trade-off, and it abstracts from other reasons for diversification such as risk sharing, market power, tax advantages, or managerial empire building. It also assumes a specific kind of contractual incompleteness: compensation can depend on implementation, but not on effort or idea quality.

11.6 Takeaway

A narrow strategy can be valuable because it gives up future flexibility to create present incentives. The firm’s refusal to pursue attractive opportunities can be rational when that refusal convinces employees that their core innovations will not be displaced.